

Commissioning & QA Sign-Off Checklist

Execute in order. Each phase gates the next; do not proceed past a failed check. References: firmware CEL-MPT-FW-002 REV 2.0, drawings CEL-MPT-DWG-001/002, install packet CEL-MPT-INS-001.

Unit / Trailer ID:		Technician:	
Firmware REV:		Date:	
Enclosure ID (ENC-01):		WiFi SSID (name only):	

Phase 1 - Bench verification (Serial only)

	Check	Pass criteria
■	Board + libraries: UNO R4 board package, Adafruit BME280 + dependencies installed	Sketch compiles with no errors
■	Bench wiring per DWG-001; BME280 on Qwiic	Visual match to drawing; LED polarity correct (220 ohm in series)
■	Serial link at 115200	Status block prints every 2 s, no garbage characters
■	BME280 detect	Serial reports found at 0x76 or 0x77; plausible temp/RH
■	Breath test on BME280	Humidity rises noticeably, then recovers
■	Door contact test	Separate magnet: OPEN + red LED; rejoin: CLOSED + green LED
■	Leak sensor test	Wet bridge: WATER DETECTED + red LED; dry: clears. If inverted, flip WATER_DETECTED_STATE and retest
■	Glitch filter sanity	Brief (<1 cycle) contact flicker does NOT flip state; sustained change does after CONFIRM_COUNT samples
■	Sensor-fault recovery	Unplug Qwiic: SENSOR FAULT + alert; replug: recovers without reboot

Phase 2 - Arduino Cloud provisioning

	Check	Pass criteria
■	Device provisioned (ECC608 bound) via Devices > Add device	Device shows online in Arduino Cloud
■	Thing variables match dashboard-fields.md exactly	temperatureF, humidityPercent (Float); waterDetected, doorOpen, systemAlert (Bool); all READ
■	Telemetry policies	Floats on 30 s time policy; booleans ON_CHANGE
■	WiFi entered in Thing Network panel	No credentials in sketch; arduino_secrets.h absent from repo
■	Cloud sketch uploaded (REV 2.0 logic in generated thingProperties)	Serial shows Cloud: CONNECTED
■	Dashboard built (2 gauges + 3 state widgets)	Live values track Serial within one publish interval
■	Offline-first check	Disable WiFi AP briefly: local LEDs + Serial keep alerting; reconnects automatically

Phase 3 - Field installation (per DWG-002 / INS-001)

	Check	Pass criteria
■	Enclosure: polycarbonate NEMA 4X, vibration isolators, bottom-entry glands with drip loops	No metal enclosure; isolators under all feet
■	Home runs: 22/4 shielded, labeled DR-01 / WL-01 / PWR-01 both ends	Labels match pull schedule; service loops at both ends
■	Continuity: each conductor end-to-end	< 2 ohm per conductor; record values
■	Shield isolation: drain to GND bar at ENC-01 ONLY	Sensor-end drain cut + taped; ohmmeter confirms no continuity sensor-end drain to chassis/ground
■	Door line pull-up: 4.7k from D2 terminal to 5V	Measured 4.3-5.1k in circuit (board unpowered)
■	Terminations on DIN blocks per internal map; ferrules on stranded	Tug test passes; no strand splay
■	Power: dedicated 5V USB-C ≥ 2 A via IP67 bulkhead	No brownout/reset during WiFi transmit bursts
■	Sensor placement	Leak sensor at bay low point; reed gap within module spec; BME280 mid-height away from heat

Phase 4 - System functional test (installed)

	Check	Pass criteria
■	Closed-lid RF check	With lid latched, Cloud stays CONNECTED ≥ 10 min; dashboard updates on schedule
■	End-to-end door test at trailer	Open/close door: dashboard state flips within ~2-5 s
■	End-to-end leak test at drip pan	Wet probe at installed location: alert on dashboard; dries + clears
■	Alert roll-up	Any single alarm drives systemAlert TRUE + red LED; all clear returns green
■	Liveness heartbeat	Thing last-update never older than ~60 s over a 15 min watch
■	Vibration shakedown (optional but recommended)	Short tow or door-slam test: no false alerts, no resets

Acceptance & sign-off

All phases passed with no open items. Unit is accepted into service as configured above.

Technician signature:		Date:	
Reviewed by (Helios Prime):		Date:	
Open items / notes:			

Record continuity readings, RSSI (if diagnostic enabled), and any deviations on the notes line. File the completed sheet with the unit's doc package.